

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

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Project: End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

This project was developed to help the business make better inventory and sales planning decisions using historical sales data. The system analyzes past sales trends, predicts future demand for the next three months, identifies unusual sales events, and groups products based on their demand patterns. These insights enable the business to optimize inventory levels, reduce excess stock, minimize stock shortages, and improve overall operational efficiency. An interactive dashboard was also developed to allow business users to monitor sales performance and forecasts without requiring any technical knowledge.

Key Findings from Exploratory Data Analysis and Forecasting

Sales Performance

- Total retail sales exceeded **\$2.26 million** during the analysis period.
- Sales showed a clear upward trend over the four-year period, indicating overall business growth.
- Sales consistently increased during the final quarter of each year, demonstrating strong seasonal demand.
- Office Supplies generated the highest forecasted sales among all product categories.
- The East region demonstrated stronger future growth compared to the West region.

Forecasting Results

Three forecasting models were developed and evaluated:

Model	MAE	RMSE	MAPE
SARIMA	20,580.71	22,190.93	21.94%
Prophet	20,250.79	22,318.41	21.86%
XGBoost	15,009.50	20,566.48	14.10%

Among the three models, **XGBoost delivered the most accurate predictions**, producing the lowest forecasting errors across all evaluation metrics. Therefore, it was selected as the production forecasting model.

Three-Month Sales Forecast

Using the XGBoost forecasting model, expected sales for the next three months are:

Forecast Month	Predicted Sales
Month 1	≈ \$83,238
Month 2	≈ \$83,061
Month 3	≈ \$87,393

Forecast Confidence

The forecast indicates that monthly sales are expected to remain relatively stable at approximately **\$83,000–\$87,000** over the next three months. While actual sales may vary due to promotions, seasonal events, or unexpected market conditions, the overall outlook suggests continued steady demand with a slight increase by the third month.

Top Three Sales Anomalies

The anomaly detection models identified several unusual sales events. The three most significant are:

Date	Weekly Sales	Likely Business Explanation
22 March 2015	\$37,703.67	Major promotional campaign or seasonal sales event.
18 November 2018	\$30,572.45	Increased festive shopping activity before the holiday season.
2 December 2018	\$35,998.90	Peak holiday demand resulting in exceptionally high sales.

These anomalies represent genuine business events rather than data errors and provide valuable insights for future inventory planning.

Product Demand Segmentation

K-Means clustering grouped products into four demand segments.

Demand Segment	Characteristics	Recommended Stocking Strategy
High Volume, Stable Demand	Strong and consistent sales	Maintain higher inventory levels and prioritize replenishment.
Growing Demand	Rapidly increasing sales	Gradually increase inventory and maintain additional safety stock.
Low Volume, High Volatility	Irregular purchasing patterns	Maintain conservative inventory and replenish based on demand.
Declining Demand	Decreasing customer demand	Reduce inventory levels and avoid overstocking.

This segmentation enables the business to allocate inventory more efficiently across different product categories.

Business Recommendations

1. Prioritize Inventory for High-Demand Products

The forecasting model predicts continued strong demand for high-performing product categories, particularly **Office Supplies** and selected Technology products. Maintaining sufficient inventory for these products will reduce stockouts and improve customer satisfaction.

2. Increase Inventory Before Seasonal Demand Peaks

Historical analysis shows that sales consistently increase during the final months of the year. Increasing inventory before these seasonal periods will help capture additional revenue and avoid lost sales caused by insufficient stock availability.

3. Use Anomaly Detection for Operational Monitoring

The anomaly detection system successfully identified unusual sales spikes associated with promotional campaigns and holiday periods. Monitoring these anomalies regularly can help management quickly identify exceptional demand, investigate unexpected changes, and improve future promotional planning.

Risk and Limitation

The forecasting models rely entirely on historical sales data. Unexpected external factors such as economic downturns, competitor actions, supply chain disruptions, changes in customer preferences, or new product launches may significantly affect future sales and reduce forecasting accuracy. Therefore, forecasts should be updated regularly using the latest available sales data.

Conclusion

The End-to-End Sales Forecasting & Demand Intelligence System provides an integrated solution for retail sales forecasting, anomaly detection, and product demand segmentation. The selected XGBoost model achieved the highest forecasting accuracy, while clustering and anomaly detection generated valuable business insights for inventory planning. The interactive Streamlit dashboard enables business users to access these insights easily, supporting informed decision-making across supply chain and finance operations.